Building a Server from Scratch

Part 4: Pages Full of Data

Let's learn how to set up a Web and database server.

Yes, we are finally coming to the juicy parts. We are now ready to create our Web server, and complement it with a database (DB) server so that we can install our favourite CMS on it. So let's jump right in.

- Role: Web server
- Priority: Required
- Dependencies: Basic server, virtualisation
- Features: Two (or three) VMs, one for the Web and one for the database
- Extensible: Yes
- Ease of setting up: Very easy but needs a lot of patience
- Software set: Pound reverse proxy, MySQL/PgSQL database server, LightTPD HTTP server, interpreters for Perl, PHP and Python2
- Set-up time: Two hours

Time for some theory

Ideally we ought to take the system offline, but this is not possible since we need to fetch software from the Internet.

By now, you will have realised that with the networking set-up that we are currently using, we'll never get VMs to talk to each other. We need to change things around a bit. One solution would be to use a second network interface on all our VMs configured for internal networking. However, this is a little dirty, because there are no DNS or DHCP servers running on this interface.

The second option would be to use a Virtual Hub, or a TAP/TUN interface to bring all the VMs together, then bridge the TAP interface to eth1 (our internal interface). So all the machines in the office can access the servers. Note that we will not bridge it to eth0. *What? Then no one can access our Web server!* Relax. We will set up a reverse proxy on the server to forward all Web requests to the Web server. Our database server need not be accessible to the outside world.

Our main server will use port forwarding (between the PC and the VM, so I can give you clear instructions this time, don't worry). Any other service, such as MediaStreaming, Jabber,